Shape and Space 2 — Pythagoras' theorem · Circle theorems

Calculator allowed. Give lengths to 3 s.f. unless exact. For every angle, give the circle theorem or angle fact you used. Marks in brackets. ★ = harder question.

A · PYTHAGORAS' THEOREM

1 In each right-angled triangle, find the missing side. [4]

a The two shorter sides are 9 cm and 12 cm. Find the hypotenuse.

b The hypotenuse is 13 cm and one shorter side is 5 cm. Find the other side.

> **a** $c^2 = 9^2 + 12^2 = 81 + 144 = 225$, so $c = 15$ cm.
> **b** $b^2 = 13^2 - 5^2 = 169 - 25 = 144$, so $b = 12$ cm (subtract when finding a shorter side).

2 A ramp 3 m long leads up to a step 30 cm high. How far from the step does the ramp start? Give your answer in metres to the nearest centimetre. [3]

> Same units: 30 cm $= 0.3$ m. The ramp is the hypotenuse:
> $x^2 = 3^2 - 0.3^2 = 9 - 0.09 = 8.91$, so $x = \sqrt{8.91} = 2.985\ldots \approx 2.98$ m.

3 Is a triangle with sides 7 cm, 24 cm and 25 cm right-angled? Show how you decide. [2]

> $7^2 + 24^2 = 49 + 576 = 625 = 25^2$. The longest side squared equals the sum of the other two squared, so by the converse of Pythagoras it **is** right-angled (the right angle is opposite the 25 cm side).

4 Find the distance between the points $A(1,\ 2)$ and $B(7,\ 10)$. [3]

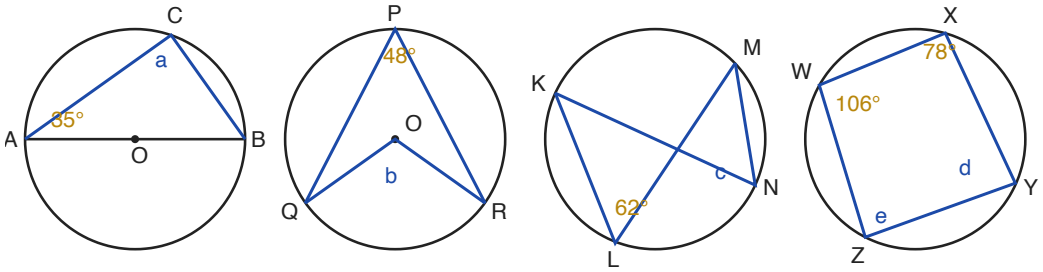
> Horizontal change $7 - 1 = 6$, vertical change $10 - 2 = 8$ — the legs of a right-angled triangle. $AB = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$.

5 A 6.5 m ladder leans against a wall and reaches 6 m up it. How far is the foot of the ladder from the wall? [2]

> $d^2 = 6.5^2 - 6^2 = 42.25 - 36 = 6.25$, so $d = 2.5$ m.

B · CIRCLE THEOREMS

6 In each diagram O is the centre of the circle. Work out the angles marked with letters, giving a reason for each. [9]



Left: AB is a diameter, angle $CAB = 35°$; find a (angle ABC). Second: angle $QPR = 48°$; find b (angle QOR). Third: chords KN and LM; angle $KLM = 62°$; find c (angle KNM). Right: cyclic quadrilateral $WXYZ$ with $W = 106°$, $X = 78°$; find d (angle Y) and e (angle Z).

> Left: angle $ACB = 90°$ (**angle in a semicircle**), so $a = 180 - 90 - 35 = 55°$ (angle sum of a triangle).
>
> Second: $b = 2 \times 48 = 96°$ (**angle at the centre is twice the angle at the circumference** on the same arc QR).
>
> Third: $c = 62°$ (**angles in the same segment** are equal — both stand on chord KM).
>
> Right: $d = 180 - 106 = 74°$ and $e = 180 - 78 = 102°$ (**opposite angles of a cyclic quadrilateral** add up to $180°$).

7 A and B lie on a circle with centre O, and angle $AOB = 130°$. C is a point on the major arc. [3]

a Work out angle ACB.

b Work out angle OAB.

> **a** Angle at the centre is twice the angle at the circumference: $ACB = 65°$.
>
> **b** $OA = OB$ (radii), so triangle OAB is isosceles: $OAB = \dfrac{180 - 130}{2} = 25°$.

8 Find the area of an equilateral triangle with perimeter 30 cm. [3]

> Side $= 10$ cm. The height splits it into two right-angled triangles with hypotenuse 10 and base 5: $h = \sqrt{10^2 - 5^2} = \sqrt{75} = 8.66$ cm. Area $= \frac{1}{2} \times 10 \times 8.66 = 43.3$ cm² (3 s.f.).

Total: 29 marks

Check: hypotenuse is opposite the right angle — add for it, subtract for a shorter side · same units before squaring · quote the theorem by name: semicircle, centre = 2 × circumference, same segment, cyclic quadrilateral.